How SLANT measures thermospheric composition

SLANT: Solar Limb Absorption of Neutrals in the Thermosphere

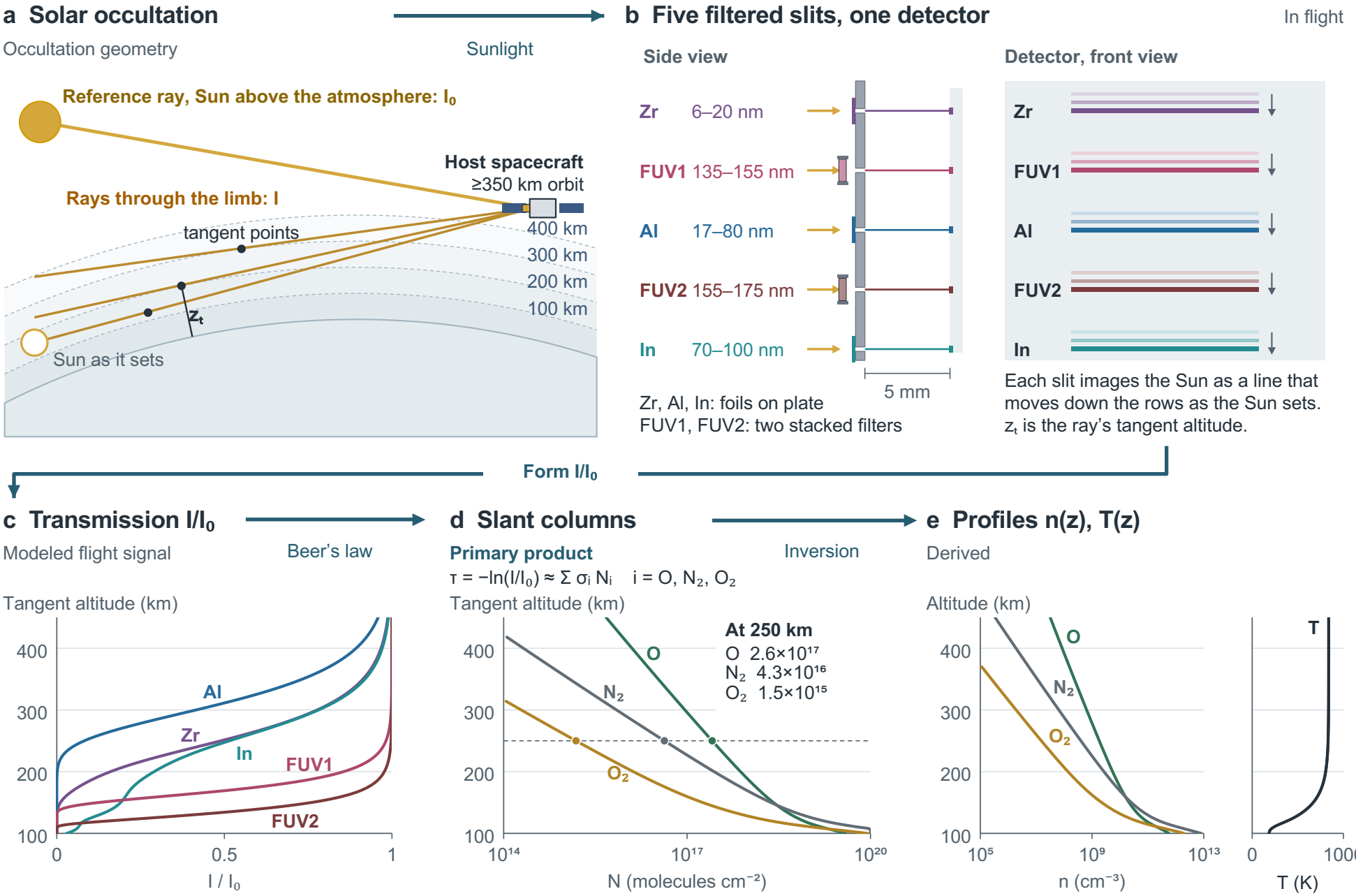
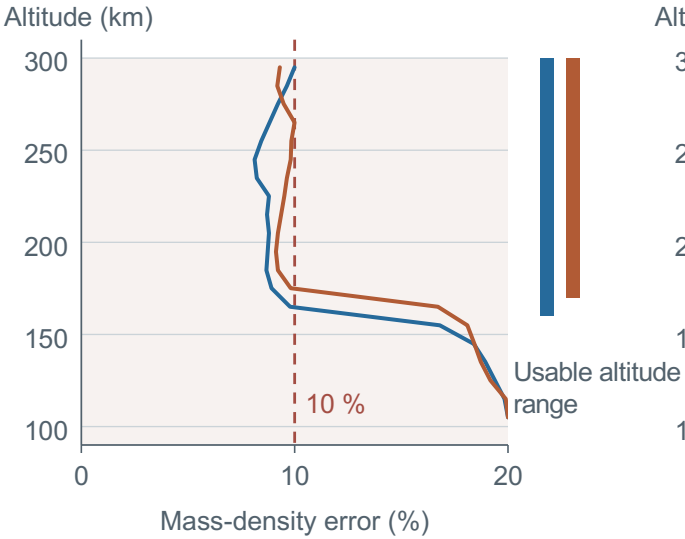


Figure 1. I/I_0 gives optical depth; Beer's law converts it to O, N_2 and O_2 slant columns, the primary product.
Inversion gives density and temperature. Curves in (c–e) model a quiet NRLMSIS 2.1 atmosphere; Zr and In use measured filter curves.

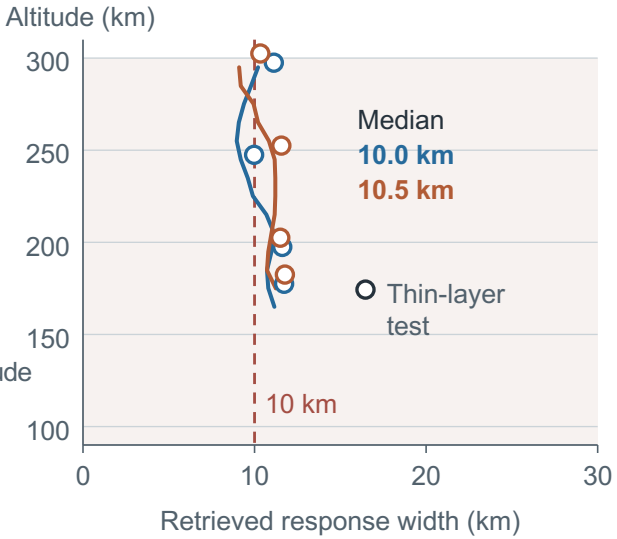
Modeled mass-density and temperature performance

One unit, one occultation — Quiet (ap 4) — Severe storm (ap 400) — Benchmark

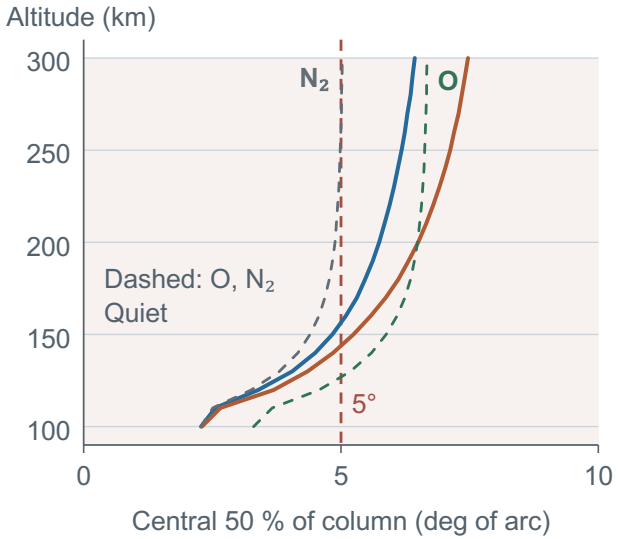
a Random error



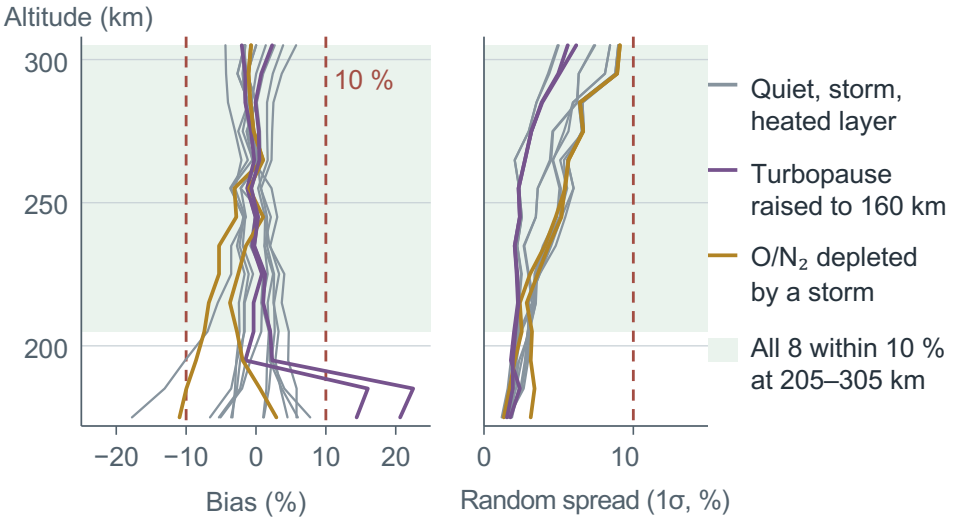
b Vertical resolution



c Horizontal resolution



d Temperature, eight test atmospheres



e Remote-sensing benchmark

Metric	Benchmark	Quiet	Severe storm
Altitude range	90–300 km	160–300 km	170–300 km
Random error	10 %	≤10 %	≤10 %
O / N_2 at 200 km	10 %	13 % / 16 %	13 % / 19 %
Vertical	10 km	10.0 km	10.5 km
Horizontal	5°	5.1–6.4°	5.9–7.5°
Retrieval bias		≤7.5 %†	≤9.8 %†
Calibration		≤2.5 % budget	≤2.5 % budget
Accuracy, total	**10 %**	**7.9 %**	**10.1 %**
Temperature	10 %, 10 km	≤10 %, 10.8 km‡	≤10 %, 10.8 km‡

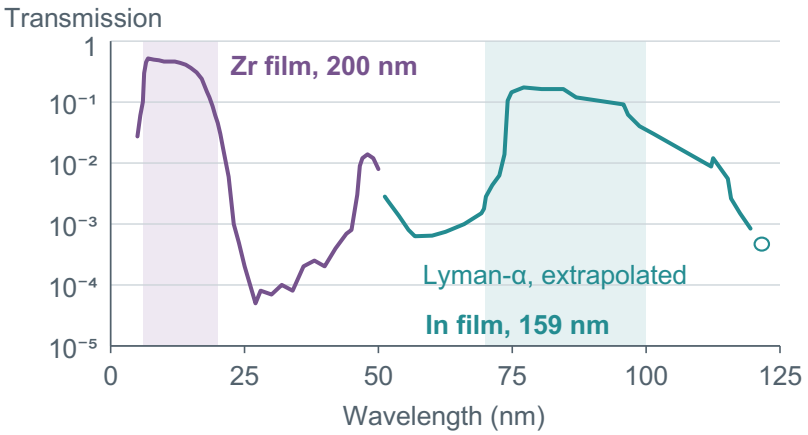
† Bias: 7 of 8 atmospheres, 170–300 km; prior wrong by its stated error.
Raised turbopause: −11 to −16% bias below 182 km.
Total = root-sum-square of bias and ≤2.5% calibration.
‡ Temperature: bias and 1σ spread, 205–305 km.
O_2 has no row: above 200 km it follows the prior.

Figure 2. Modeled mass density uses measured Zr and In curves, blur and noise. Usable: ≤10% random error, and a true change recovered to within 20%. (d) Temperature: eight atmospheres, 30 noise draws each; priors vary O and N_2 by 45% and O_2 by 60%. Shading marks benchmark altitudes.

Component evidence and the calibration plan

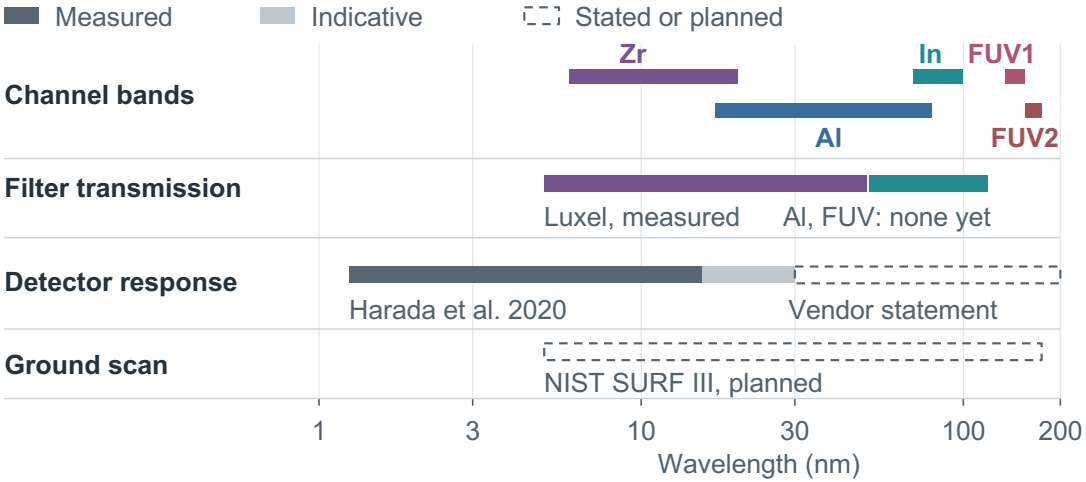
Measured component data, calibration plan, and bench results

a Measured filter transmission



Shading: nominal channel bands. Data: Luxel Corp.

b Where measured evidence exists



c Calibration plan

1 Every occultation

Reference rows above the atmosphere give I_0.
Stable common gain and brightness cancel in I/I_0.
Flat-field drift: a 1 % random term. Flares reject a pass.

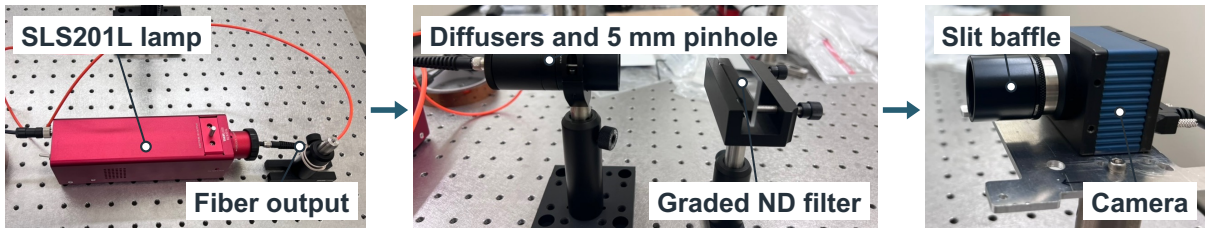
2 Every occultation

Frames below 105 km, where EUV channels see no in-band light, measure the out-of-band leak so it can be subtracted.

3 On the ground, before launch

NIST SURF III, 5–175 nm, measures the engineering unit's response and a witness filter's optical depth; the flight films share its Luxel run.

d Visible-light bench test, Carthage



Flight and bench optics

	Flight	Bench
Source angle	9.3 mrad	10.6 mrad
Slit width	25 µm	100 µm
Slit to detector	5.0 mm	17.5 mm
Blur ÷ slit width	**1.86**	**1.86**
Fresnel number	**0.71–20.8**	**0.74–1.02**
FUV Fresnel	**0.71–0.93**	**0.74–1.02**
Wavelength	6–175 nm	580–800 nm

Bench tests

1 **Optical depth from I/I_0**
Measured $|\Delta\tau| \leq 0.0096$ for $\tau = 0.6–2.25$.

2 **Image position over 10 min**
Position drift ≤0.05 detector row.

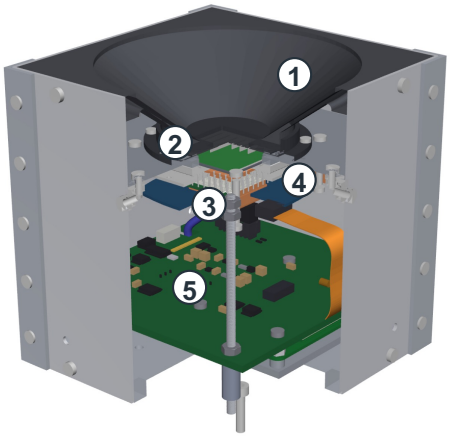
3 **Two absorbers at once**
Recover mixture from separate responses.

Figure 3. (a) Measured Zr and In transmission; the Lyman-α point is extrapolated. (b) Evidence coverage. (c) In-flight corrections and planned synchrotron calibration. (d) The visible-light bench test matches the blur-to-slit ratio and overlaps the FUV diffraction regime; it does not reproduce the full EUV range. Test 1 is measured against a 0.02 limit; tests 2 and 3 are targets.

The payload and how it fits a host

Payload architecture, host requirements, and flight heritage

a SLANT payload



1 Baffle, matte black

2 Slit plates and five filters

3 Sensor and heat path

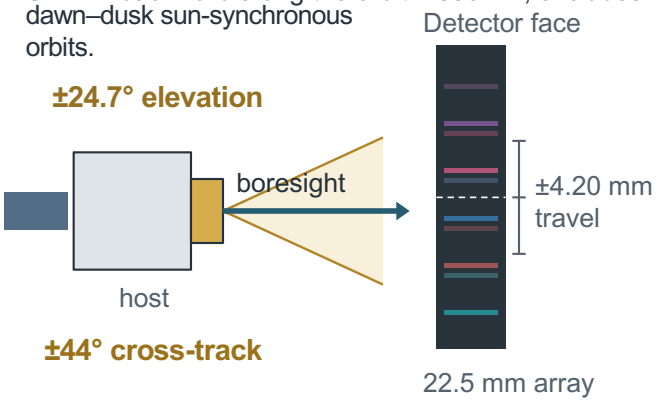
4 XIMEA sensor electronics

5 CM5 on carrier board

103 × 103 × 107 mm; modeled mass ≈665 g

b Acceptance cone

Three-axis-stabilized spacecraft; fixed bracket points SLANT backward along the orbit. ≥350 km; excludes dawn–dusk sun-synchronous orbits.



2.76 mm sweep; ±51.4° cross-track alone.
At combined tilts, mask any clipped rows.

c Signal and host interfaces

Optical head

Slit plates, filters; sensor on the XIMEA sensor board

↓ 60-pin link

XIMEA controller

↓ USB 3

Carthage carrier board with CM5

Row sums, storage, host link; power switching, watchdog

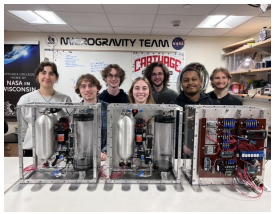
↓ Power: Glenair 801
↓ Data: 15-way micro-D

Host spacecraft

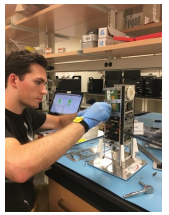
d Host requirements

Mounting	4 × M3 on a 66 mm square; conductive base; 800 g allocated. Mounting tolerance: ±24.7° elevation, ±44° cross-track. Keep ±40° × ±44° free of obstructions.
Thermal	Conductive heat path required. Baseplate −20 to +50 °C for survival, +20 °C or colder while acquiring. Detector uncooled.
Electrical	9–36 V, isolated; 5.7–9.7 W acquiring; 11.6 W allocated; off between passes; 0.33–0.55 W orbit average. Host switches power at times SLANT reports, or stays on.
Data and time	CAN 2.0B or RS-422; 7.2 MB/day (0.67 kbit/s mean); 1 PPS and UTC to 10 ms; position to 100 m; attitude knowledge 0.5°
Delivered	Interface control document: mounting, thermal, power, data, timing, keep-out; test reports; flight and ground software.
Not needed	Pointing service, radio, GNSS receiver, active cooling.

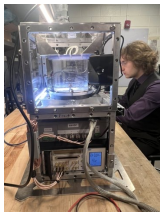
e Carthage payloads



NASA UPI Challenge: three identical payloads



CaNOP camera, engineering model



MUTT refueling, TechLeap award

22	2	6	5	2	1
Parabolic campaigns	ISS experiments	Suborbital flights	Sounding rockets	CubeSat missions	IM-3 lunar instrument

Host computer: Raspberry Pi CM5

Its predecessor, CM4, flies on SatGus and VERTECS.

Figure 4. (a) CAD layout and modeled mass. (b) Mounting tolerance: ±24.7° elevation and ±44° cross-track together, ±51.4° cross-track alone. (c–d) Host interfaces and requirements; allocations add 20 % to the CAD mass and estimated power. (e) Team flight heritage; flight heritage of compute board.

Production and network coverage

Production, simulated coverage, and projected network-mean error

a Production flow

Suppliers · per production lot

Slit sheet and carrier
Luxel builds

Metal foils: Zr, Al, In
Luxel bonds them to the sheet

FUV filters
Catalogue vendor, per order

Detectors
Sensor vendor, per wafer lot

Boards
Fabrication and assembly, per panel

Carthage · every unit

Optical head
Fit partitions, set the 5.0 mm standoff, clamp FUV filters

Unit assembly
Boards, heat path, housing

Unit test
Flat field, dark current against temperature, timing, vibration

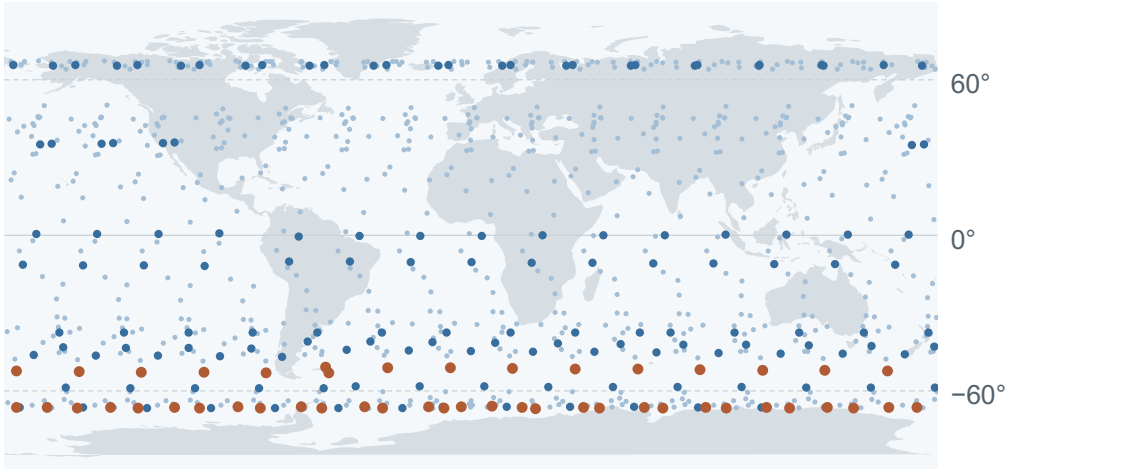
Calibration

Ground calibration
Angular scan: every slit plate
NIST SURF III, 5–175 nm: one unit per 20-unit lot

Launch on a host

In flight, every occultation
Reference rows give I_0;
units are cross-compared

b Simulated occultations, 1 January 2026

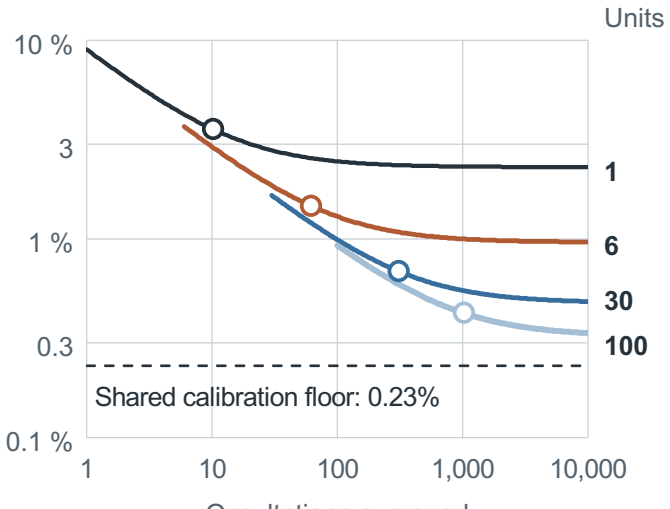


● Units 1–6: 47 total ● Units 7–30: 189 total ● Units 31–100: 940 total

Model: independent 8.7% random errors per occultation.
Assumed 2.3% unit offset; 1% of its variance is shared.
Balanced sampling of a common density; physical variability excluded.

c Random + calibration error

○ One day at annual-average rate



Mean mass density at 200 km, severe storm

Figure 5. (a) Batch production and unit testing. (b) Sunset rays at 200 km; hosts: 42 in the 53° shell, 25 sun-synchronous, 33 other; counts cumulative. (c) Network-mean error; page 2 bias adds. Markers: one day at 10.33 occultations per unit. With 30 % of each offset shared per 20-unit lot, 100 units reach 0.64 %.